

Quantum Mechanics – II

Assignment – III

Due – 15/05/2025

1.

- (a) *Spatial wavefunctions:* Let $|\alpha\rangle, |\beta\rangle$ be two orthogonal single-particle states for particles in 1-d. with $\psi_\alpha(x) = \langle x|\alpha\rangle$ and $\psi_\beta(x) = \langle x|\beta\rangle$. Define the distinguishable, symmetric and antisymmetric states to be

$$|\Psi_D\rangle \equiv |\alpha\rangle \otimes |\beta\rangle \quad (1)$$

$$|\Psi_S\rangle \equiv \frac{|\alpha\rangle \otimes |\beta\rangle + |\beta\rangle \otimes |\alpha\rangle}{\sqrt{2}} \quad (2)$$

$$|\Psi_A\rangle \equiv \frac{|\alpha\rangle \otimes |\beta\rangle - |\beta\rangle \otimes |\alpha\rangle}{\sqrt{2}} \quad (3)$$

Using tensor product notation, the position of the first (resp. second) particle is given by the operator $\hat{x}_1 \equiv \hat{x} \otimes I$ (resp. $\hat{x}_2 \equiv I \otimes \hat{x}$). Define $D_D = \langle \Psi_D | (\hat{x}_1 - \hat{x}_2)^2 | \Psi_D \rangle$, $D_S = \langle \Psi_S | (\hat{x}_1 - \hat{x}_2)^2 | \Psi_S \rangle$ and $D_A = \langle \Psi_A | (\hat{x}_1 - \hat{x}_2)^2 | \Psi_A \rangle$. Calculate $D_S - D_D$ and $D_A - D_D$. Give a brief intuitive explanation of the signs of your answers.

- (b) *Spins:* Now consider two spin- s particles. For a single particle, let $|s, m_a\rangle$ be the $\hbar m_a$ eigenstate of the S_z operator. Suppose $m_a \neq m_b$. Define

$$|\Psi_D\rangle \equiv |s, m_a\rangle \otimes |s, m_b\rangle \quad D_D \equiv \langle \Psi_D | (\vec{S}_1 - \vec{S}_2)^2 | \Psi_D \rangle \quad (4)$$

$$|\Psi_S\rangle \equiv \frac{|s, m_a\rangle \otimes |s, m_b\rangle + |s, m_b\rangle \otimes |s, m_a\rangle}{\sqrt{2}} \quad D_S \equiv \langle \Psi_S | (\vec{S}_1 - \vec{S}_2)^2 | \Psi_S \rangle \quad (5)$$

$$|\Psi_A\rangle \equiv \frac{|s, m_a\rangle \otimes |s, m_b\rangle - |s, m_b\rangle \otimes |s, m_a\rangle}{\sqrt{2}} \quad D_A \equiv \langle \Psi_A | (\vec{S}_1 - \vec{S}_2)^2 | \Psi_A \rangle \quad (6)$$

Calculate D_D, D_S, D_A in terms of m_a, m_b . Order them from smallest to largest; i.e. write down an expression of the form $D_X \leq D_Y \leq D_Z$ with $\{X, Y, Z\}$ some permutation of $\{D, S, A\}$.

[*Hint:* You will find it useful to express $\vec{S}_1 \cdot \vec{S}_2$ in terms of the appropriate raising and lowering operators.]

2.

Let ρ be a finite-dimensional density matrix. Recall that ρ is said to be a pure state if $\rho = |\psi\rangle\langle\psi|$ for some $|\psi\rangle$. Prove that $\text{tr}(\rho^2) = 1$ if and only if ρ is pure.

3.

Write down three spin-1/2 states $|\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle$ such that if each occurs with probability $1/3$, the resulting density operator is $\frac{I}{2}$.

4.

Consider an electron spin in the state

$$\rho = \begin{pmatrix} \rho_{++} & \rho_{+-} \\ \rho_{-+} & \rho_{--} \end{pmatrix}$$

that experiences a magnetic field $B\hat{z}$. The Hamiltonian is then $H = -\gamma B\hat{S}_z$ with $\gamma = g_e e/2m_e$. Suppose that the field strength B is drawn from a Gaussian distribution with mean 0 and variance σ^2 ; i.e. the probability density of B is

$$f(B) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{B^2}{2\sigma^2}}.$$

Let ρ' be the state that results from applying this field for time t and averaging over the possible values of B . Compute ρ' .

5.

Consider a system made up of two spin $\frac{1}{2}$ particles. Observer A specializes in measuring the spin components of one of the particles (s_{1z}, s_{1x} and so on), while observer B measures the spin components of the other particle. Suppose the system is known to be in a spin-singlet state, that is, $S_{\text{total}} = 0$.

- What is the probability for observer A to obtain $s_{1z} = \hbar/2$ when observer B makes no measurement? Same problem for $s_{1x} = \hbar/2$.
- Observer B determines the spin of particle 2 to be in the $s_{2z} = \hbar/2$ state with certainty. What can we then conclude about the outcome of observer A's measurement if (i) A measures s_{1z} and (ii) A measures s_{1x} ? Justify your answer.

6.

It is obvious that two nonidentical spin 1 particles with no orbital angular momenta (that is, s -states for both) can form $j = 0$, $j = 1$, and $j = 2$. Suppose, however, that the two particles are *identical*. What restrictions do we get?

- Work out the details of a teleportation and protocol using the 3-qubit GHZ state.

8.

The von Neumann entropy of a system described by a density matrix ρ is defined as

$$S = -\text{Tr} \rho \ln \rho$$

Prove the following properties of the von Neumann entropy of a quantum system.

- (a) The von Neumann entropy is invariant under unitary transformations of the density matrix ρ .
- (b) If ρ has D non-vanishing eigenvalues then

$$S(\rho) \leq \log D$$

with equality only when all the non-zero eigenvalues are equal. This means that the entropy is maximized when the quantum state is chosen randomly.

- (c) If ρ_1 and ρ_2 describe two density matrices for a system, then first show that $\rho(\lambda) = \lambda\rho_1 + (1 - \lambda)\rho_2$ is also a legitimate density matrix. Next show that

$$S(\rho(\lambda)) \geq \lambda S(\rho_1) + (1 - \lambda)S(\rho_2)$$

This means that the more ignorant we are about how the state was prepared the higher the von Neumann entropy.

- (d) For a composite system $A + B$ in a general mixed state described by density matrix ρ_{A+B} ,

$$S(\rho_{A+B}) \leq S(\rho_A) + S(\rho_B)$$

with equality for uncorrelated systems where the full density matrix factorizes.

9.

Show that for an operator a which, with its adjoint, obeys the anticommutation relation $\{a, a^\dagger\} = aa^\dagger + a^\dagger a = 1$, then the operator $N = a^\dagger a$ has eigenstates with the eigenvalues 0 and 1.

10.

Three identical particles are in a one-dimensional harmonic oscillator potential well with classical angular frequency ω .

- a. Write the complete time-independent Hamiltonian for this system, and express it in coordinate space as a differential equation whose solution is the three-body wave function $\Psi(x_1, x_2, x_3)$.
- b. Assume the particles have zero spin. Use the single-particle wave functions to construct the ground-state wave function $\Psi_0(x_1, x_2, x_3)$, and show that it satisfies the differential equation in (a), and find the ground-state energy.
- c. Assume the particles have spin $\frac{1}{2}$. Repeat (b), and also construct the ground state spin state from single-particle spin eigenstates.